What is a mineral?

Minerals are the substances that make up the Earth's rocks. Each one has its own unique chemical composition and internal atomic structure – indeed, a mineral is defined by its chemical elements and by the atomic structure of its crystallization. Minerals are usually formed by inorganic processes, although there are organically produced substances such as the hydroxylapatite in teeth and bones that are also considered minerals. Certain substances, including opal and glass, resemble minerals in appearance, chemistry, and occurrence, but do not have a regularly ordered internal arrangement and so do not exhibit crystallinity: these are known as mineraloids.

A few minerals occur as a single chemical element: these are known as "native elements" and include gold, silver, and diamond (see below). However, most minerals are chemical compounds, composed of two or more chemical elements. There are around 100 types of mineral that are considered common, out of more than 5,100 known minerals.

Mineral classification

Minerals are grouped according to their chemical composition. A mineral compound has positively and negatively charged atoms or groups of atoms: the atoms that carry the negative electrical charge determine which chemical group a mineral is assigned to. The largest mineral group, the silicates, is further divided into six sub-groups based on their different chemical structures.


Gold nugget


Fluorite crystals

Rough chrysocolla with azurite

Celestine cystals

Native elements

Minerals made up of atoms from a single element are known as native elements. The most common are metals such as copper, iron, silver, gold, and platinum, and non-metals such as sulphur and carbon (as graphite and diamond). A few others occur in minute amounts, often alloyed with other native elements.

Halides

Halides consist of various metals combined with one of the common halogen elements: fluorine, chlorine, bromine, or iodine. There are three categories of halide: simple halides; halide complexes; and oxyhydroxy-halides. All halides are soft, and thus there are few gemstone varieties, except for fluorite.

Carbonates

A mineral in the carbonate group is characterized as having a carbon atom at the centre of a triangle of oxygen atoms, which gives rise to trigonal symmetry (see pp.18–19). Examples of carbonates as gemstones include chrysocolla, calcite, smithsonite, and malachite.

Sulphates

Sulphate minerals have a crystal structure consisting of four oxygen atoms, with a sulphur atom in the centre; this combines with one or more metals or semi-metals. Some examples of sulphates include baryte, celestine, and alabaster (a variety of the sulphate mineral gypsum).